

Alakh Bathani

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EDUCATION

Class-12(Percentage-95.8) - Ashadeep International School

Apr 2022 - Mar 2024

Bachelor of Technology in Artificial Intelligence - Sardar Vallabhbhai National Institute of Technology **Aug 2024- May 2028**
(Current CGPA: 9.02)

TECHNICAL SKILLS

Languages: Modern C++17/20, C, Python, JavaScript, SQL

Frameworks & Libraries: C++ STL, React.js, Node.js, Express.js, Pandas, NumPy, Matplotlib, Seaborn

Web Technologies: HTML5, CSS3, Tailwind CSS, RESTful APIs

Databases & Cloud: MongoDB, PostgreSQL, Firebase

Machine Learning: Scikit-Learn, Supervised Learning, Decision Trees, Random Forest, AdaBoost, Gradient Boosting, XGBoost, Cross Validation

Core Concepts: Data Structures & Algorithms(in C++), Exploratory Data Analysis (EDA), Feature Engineering, Database Management Systems(DBMS), OOP, Computer Organization(CO), Operating Systems(OS)

PROJECTS

Distraction Free Adaptive Reader

Next.js, TypeScript, AWS Bedrock, PostgreSQL, Tailwind CSS

- **Architected** an implicit telemetry engine in React and TypeScript to track scroll velocity, dwell time, and text selection for real-time reading friction analysis.
- **Integrated** AWS Bedrock LLMs through Next.js serverless API routes to dynamically adapt reading content by injecting contextual synonyms during short pauses and translating paragraphs after prolonged reading stalls.
- **Optimized** backend synchronization by implementing a dual-throttled persistence layer that debounced rapid local state updates and batched PostgreSQL writes every 5 seconds, eliminating network bottlenecks and improving application stability.
- **Built** a post-session analytics pipeline to extract difficult vocabulary and leveraged LLMs to automatically generate categorized Concept Graphs, enabling targeted review and personalized learning.

AI-Native Autonomous Testimonial Collection

React, Gemini AI, Deepgram, Cloudinary, Firebase

- **Engineered** a full-stack Next.js web application that automates user testimonial collection, featuring an interactive AI host that conducts browser-based video interviews.
- **Constructed** a real-time conversational engine by integrating the Google Gemini API for dynamic follow-up question generation and Deepgram for low-latency Speech-to-Text transcription.
- **Implemented** complex browser media capture using the MediaRecorder API to multiplex system audio and microphone streams, featuring custom Voice Activity Detection (VAD) via AudioContext to track precise silence thresholds.
- **Created** a responsive, lip-synced UI avatar using React and the Web Speech API, generating real-time viseme animations and facial expressions based on contextual sentiment analysis.
- **Embedded** a robust cloud media storage solution using Firebase and Cloudinary for secure video blob uploading, connected to the Vizard API to automatically convert raw interviews into captioned, vertical-format social media reels.

POSITIONS OF RESPONSIBILITY

Tech Head:

Cutting Edge Visionaries, NIT Surat

May 2026 - Present

Co-Head:

Mindbend - Gujarat's Largest Techno-Managerial Fest, NIT Surat

Nov 2025 - Present

Executive:

Association for Computing Machinery, NIT Surat

Sep 2025 - Present

ACHIEVEMENTS

- **Runner-Up – PowerMind Hackathon:** Achieved 2nd place among competing teams at CDC SVNIT × Adani Group's PowerMind Hackathon by developing an AI-powered electricity forecasting system featuring explainable XGBoost models, efficient data processing, and advanced feature engineering.
- **Runner-Up – Enfinity Hackathon:** Secured runner up position among multiple competing teams by developing an innovative and scalable solution under tight time constraints.
- **Finalist – Odoo Hackathon:** Built a scalable supply-chain optimization module within 24 hours.
- **Finalist – Hack The Tank 3.0 Hackathon:** Obtained finalist position by designing and presenting prototype under competitive hackathon conditions.